Evidential Multi-Agent Orchestration for Multimodal Urban Flood Risk Assessment: A Subjective Logic Framework with Provenance-Aware Explanations

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ABSTRACT

Urban flooding is among the most economically destructive categories of natural hazard, with the February–March 2022 south-east Queensland event ranking as one of Australia's costliest flood episodes of recent decades. Effective operational decision-support requires fusing heterogeneous streams — satellite radar, multi-spectral optical imagery, reanalysis precipitation, terrain and land-cover layers, gauge and air-quality stations, and unstructured hydrological bulletins — that arrive at irregular cadences with frequent inter-source disagreement and mutual sensor outages. The downstream decision demands not only a class prediction but also a calibrated, auditable account of which sensors contributed, which were absent, and how strongly the system believes each candidate outcome.

1. Introduction

Urban flooding is among the most economically destructive categories of natural hazard, with the February–March 2022 south-east Queensland event ranking as one of Australia's costliest flood episodes of recent decades. Effective operational decision-support requires fusing heterogeneous streams — satellite radar, multi-spectral optical imagery, reanalysis precipitation, terrain and land-cover layers, gauge and air-quality stations, and unstructured hydrological bulletins — that arrive at irregular cadences with frequent inter-source disagreement and mutual sensor outages. The downstream decision (issuing an inundation warning, prioritizing evacuation, allocating sandbag resources) demands not only a class prediction but also a calibrated, auditable account of which sensors contributed, which were absent, and how strongly the system believes each candidate outcome.

Three architectural families dominate current literature. LLM-orchestrated agentic pipelines delegate sub-tasks to specialist agents and arbitrate conflicts through prompt-chained reasoning [Jiang et al., 2026]. Remote-sensing-grounded multi-agent systems wrap tool-use and retrieval around geospatial foundation models [Talemi et al., 2026, Redaelli et al., 2026]. A third tradition couples dynamic knowledge graphs with environmental inference nodes to mediate between sensors and decision outputs [Hofmeister et al., 2024]. Parallel to this agentic wave sits an evidential-operator literature — multi-source Subjective Logic fusion [van der Heijden et al., 2018], partial-observable projection, credibility-weighted trust models, and evidential deep-learning heads — that is mathematically rigorous but not yet interfaced end-to-end with multi-agent environmental pipelines.

A close reading of these families exposes four structural limitations that block operational deployment. First, LLM-mediated arbitration produces uncalibrated probabilities: LLM-arbiters attain moderate ranking quality but collapse minority-class recall, generating a confident-but-wrong majority-class skew. Second, missing modalities are silently recovered or hallucinated, with no formal monotonicity guarantee on epistemic uncertainty under sensor dropout. Third, per-agent credibility is not propagated — a historically unreliable agent is weighted identically to a historically reliable one. Fourth, no source-provenance trail exists: post-hoc feature attribution explains feature importance but not which sensor or document caused a contribution, so an analyst cannot audit which data source was stale, missing, or uncalibrated. These four limitations motivate a single framework that combines specialist-agent decomposition, formal evidential reasoning, monotonic missing-modality projection, and source-level provenance.

We propose AEGIS[1] — an open-source framework: Agentic Evidential Geographic Intelligence for Sustainability, a specialist multi-agent framework in which each of five evidential agents — Climate, Satellite, Land-Cover, Air-Quality / Sensor, and Document Intelligence — emits a formal Subjective Logic opinion over a shared four-state Dirichlet frame. A deterministic coordinator inspects inter-agent agreement via a Jensen-Shannon signal. When agents agree, it routes through Consensus & Compromise Fusion; when they disagree, it routes through Weighted Belief Fusion weighted by per-agent credibility γ. Missing modalities automatically trigger a partial-observable projection that guarantees the epistemic uncertainty u grows monotonically as modalities drop, rather than subtracting false confidence. The cell-level fused opinion is concatenated with raw multi-modal features and passed to a hybrid LightGBM evidential

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[1] https://github.com/ali-Hamza817/AEGIS

head that retains the formal uncertainty signal as an input feature while preserving a competitively accurate prediction surface. Every emitted and fused opinion is logged to a DuckDB-backed provenance ledger with manifest IDs and source references, powering an interactive Leaflet dashboard that surfaces per-cell uncertainty, per-agent credibility, and per-source contribution shares.

The remainder of this paper is organized as follows. Section 2 reviews related work across agentic environmental AI, subjective-logic foundations, and conflict-resolution techniques. Section 3 develops the mathematical foundations of AEGIS. Section 4 details the architecture. Section 5 describes the Brisbane 2022 study, datasets, baselines, and metrics. Section 6 reports hypotheses H1–H4 with a mechanistic ablation isolating the contribution of the evidential machinery. Section 7 concludes with limitations, broader applicability to other environmental risk domains, and future work.

2. Related Work

The earliest formal treatment of disagreement between distributed agents was articulated by Nielsen and Parsons [2006], who showed that Bayesian networks held by separate agents can be fused through formal argumentation schemes, introducing an explicit syntax for managing inter-agent inconsistency. Wang and Singh [2007] then proposed the canonical Brier-score trust model that underpins nearly every subsequent reputation system in MAS, demonstrating that continuous track-record scoring outperforms binary endorsement under uncertainty. Together these two contributions fixed the foundational vocabulary still in use today: agents maintain beliefs, disagreements must be reasoned about formally, and trust must be tracked quantitatively.

The limitations of purely probabilistic MAS reasoning — inability to express honest "I don't know," no formal operator for missing observations, and no algebraically closed multi-source fusion — were addressed by the Subjective Logic operator algebra. Kaplan et al. [2015] introduced the partial-observable projection that mathematically guarantees epistemic uncertainty rises when evidence is absent, and van der Heijden et al. [2018] derived the Weighted Belief Fusion and Consensus & Compromise Fusion operators that compose an arbitrary number of opinions without losing algebraic closure. These contributions supplied the missing operator algebra: a framework in which trust-weighted multi-source fusion is provably defined and missing signals propagate as quantified uncertainty rather than false confidence.

A second parallel wave applied evidential reasoning at the discriminative learning layer. Li et al. [2024] introduced adaptive uncertainty-based learning for text-based person retrieval; Liu et al. [2025] proposed MERIT for multi-view evidential liver-fibrosis staging; and Chen et al. [2025] re-examined essential versus nonessential settings of evidential deep learning. In parallel, Sun et al. [2024] introduced T2MAC, which propagates per-agent credibility through selective communication engagement, and Chikoti [2025] consolidated the conflict-resolution taxonomy within MAS. These works demonstrated that evidential reasoning could be deployed at both the symbolic-fusion layer and the discriminative learning layer, but the two traditions remained disconnected from physical environmental inference.

The first attempts to apply this machinery to physical environmental inference emerged through dynamic knowledge graphs. Hofmeister et al. [2023] demonstrated cross-domain flood-risk assessment for smart cities using a dynamic semantic substrate mediating between data sources and inference nodes. Hofmeister et al. [2024] extended this into an explicit semantic-agent framework for automated flood assessment, supplying the architectural intuition that a mediating ontology was required. The agent layer remained, however, in early form and did not yet integrate computable trust or monotonic missing-modality handling.

The agentic environmental-AI wave accelerated between 2025 and 2026. Foundational surveys catalogued the architecture space [Sayyad et al., 2025, Talemi et al., 2026], and domain-specific deployments followed rapidly across fire and flood applications: Bairam Zadeh et al. [2025] proposed a conceptual high-level MAS for wildfire management; Lee et al. [2025] built multi-agent geospatial copilots for remote-sensing workflows; Shabbir et al. [2025] benchmarked tool-augmented agents through ThinkGeo; Sandeep et al. [2026] framed context-aware MAS for wildfire insights; Palanca Cantonjos and Biswas [2026] deployed AgroAskAI for smallholder-farmer queries; Jiang et al. [2026] introduced Flood-LLM, fusing multi-source urban data for property-level flood-risk assessment; and Redaelli et al. [2026] operationalised the SaferPlaces agentic AI for global flood-risk intelligence. Every one of these systems, however, relies on prompt-chained arbitration: an LLM receives agent summaries and emits a decision. None of them delivers calibrated uncertainty, monotonic missing-modality handling, per-agent credibility, or source-level provenance.

The present paper introduces AEGIS to address exactly these four unresolved structural limitations. By interfacing the agentic architectural intuitions [Hofmeister et al., 2024, Talemi et al., 2026] with the formal Subjective Logic operator algebra [Kaplan et al., 2015, van der Heijden et al., 2018], the canonical trust model [Wang and Singh, 2007], and the credibility-aware communication work of Sun et al. [2024] and Chikoti [2025], AEGIS fuses five specialist agents over a shared Dirichlet frame, propagates calibrated epistemic uncertainty under arbitrary sensor dropout, and exposes a per-cell provenance manifest through a DuckDB-backed dashboard. We show on the Brisbane February–March 2022 case study that the resulting framework matches monolithic accuracy (F1-Macro 0.7190 vs 0.7106) while uniquely providing calibrated uncertainty, missing-modality monotonicity, and per-agent reputational weighting.

3. Mathematical Foundations

3.1. Opinion algebra and Dirichlet bijection

We model the environmental state space over the frame of discernment $\mathbb{X} = \{\theta_{\text{dry}}, \theta_{\text{saturated}}, \theta_{\text{surface}}, \theta_{\text{inundation}}\}$, a four-state classification distinguishing progressively wetter hydrological conditions. A Subjective Logic opinion over $\mathbb{X}$ is the tuple $\omega = (\mathbf{b}, u, \mathbf{a})$, where $\mathbf{b} \in \mathbb{R}^4$ is the belief mass vector over the four states, $u \in \mathbb{R}$ is the epistemic uncertainty mass, and $\mathbf{a} \in \mathbb{R}^4$ is the prior base-rate vector. The opinion satisfies the closure constraint $\sum_{k=1}^{4} b_k + u = 1$ with $b_k \geq 0$, $u \geq 0$, and $\sum_{k=1}^{4} a_k = 1$. Given Dirichlet evidence parameters $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \alpha_3, \alpha_4)$ with total evidence weight $C = \sum_{k=1}^{4} \alpha_k$ and non-informative prior constant W, the bijective mapping onto opinion space is

$$b_k = \frac{\alpha_k - C \cdot a_k}{C + W}, \quad u = \frac{W}{W + C} \tag{1}$$

and the projected expected probability over each state is

$$P(x_k) = b_k + a_k \cdot u \tag{2}$$

which reduces to a standard probability vector when $u = 0$ and to a diffuse vacuous distribution as $u \to 1$.

3.2. Partial-observable projection

When a sensor or modality yields no observation, the agent formally projects its evidence onto the unobserved dimensions rather than imputing synthetic values. Following the projection rule of Kaplan et al. [2015], the projected Dirichlet parameters are $\alpha_k^{\text{proj}} = C \cdot a_k$. This forces $b_k \to 0$ for all k and $u \to 1.0$, yielding the monotonicity property that the AEGIS framework exploits throughout:

Theorem. *When an observing agent loses one or more modalities, the projected epistemic uncertainty u is a strictly non-decreasing function of the count of missing modalities [Kaplan et al., 2015, van der Heijden et al., 2018].*

Operationally, this guarantee means missing data produces auditable quantification rather than silent hallucination — every dropout event is detectable through a monotonic u increase recorded in the provenance ledger, and is therefore distinguishable from confident-but-wrong prediction patterns.

3.3. Multi-source fusion: WBF and CCF

The coordinator composes a sequence of $N \geq 2$ agent opinions over the same frame through one of two compounding operators. The first is Weighted Belief Fusion, the credibility-weighted extension of the canonical two-source rule, applicable for N-ary aggregation when per-agent reputations are non-identity. The second is Consensus & Compromise Fusion, an averaging operator that is provably associative and commutative over the opinion lattice for any $N \geq 2$ [van der Heijden et al., 2018]. Operator selection is driven by inter-agent agreement, as detailed in Section 4. Both operators close the operator algebra over the tuple $\omega = (\mathbf{b}, u, \mathbf{a})$ and are selected such that the partial-observable projection invariant is preserved under composition.

3.4. Specialist-credibility update

Each specialist agent carries a Brier-score reputation $\gamma_i \in [\gamma_{\min}, 1.0]$ updated against the realised outcome online. For an agent emitting projected probabilities $\hat{p}_{i,k}$ against the one-hot realised state y_k, the score is

$$\text{BS}_i = \frac{1}{K} \sum_{k=1}^{4} \left(\hat{p}_{i,k} - y_k\right)^2 \tag{3}$$

and the reputation is updated as

$$\gamma_i^{(t+1)} = \text{clip}\left(\gamma_i^{(t)} + \eta \cdot (1 - 2 \cdot \text{BS}_i),\ \gamma_{\min},\ 1.0\right) \tag{4}$$

with learning rate η and lower bound $\gamma_{\min}$. The clipping step enforces a non-zero agent vote inside WBF weighting even when an agent has scored poorly over many rounds — an empirically important property so an agent cannot be silently zeroed out and remove itself from disagreement-resolution channels. This dynamic reputation tracking is what differentiates the AEGIS framework from monolithic late-fusion alternatives; an unreliable agent's effective contribution to WBF decays over successive rounds without manual re-tuning [Wang and Singh, 2007].

3.5. Hybrid-head composite feature space

The evidential prediction head consumes a 27-dimensional composite feature vector assembled through a concatenation

$$\mathbf{F} = [\, \mathbf{b} \mid u \mid \mathbf{a} \mid \text{raw features} \,] \tag{5}$$

The first twelve dimensions encode the four-state opinion tensor carried from the SL coordinator; the remaining fifteen dimensions encode the raw multi-modal feature channels used for fitting (climate, SAR polarisations, NDWI, NDVI, terrain slope, PM2.5, embedded document features). Critically, the u mass is preserved as a 13th explicit feature rather than collapsed into a scalar probability. The LightGBM gradient-boosting model consumes the full 27-dimensional vector in a single-shot fit. The composite space allows AEGIS to simultaneously inherit the calibrated uncertainty of the opinion layer, achieve accuracy competitive with raw-feature monolithic alternatives, and propagate the missing-modality invariant end-to-end. In Section 6, we empirically decompose whether the SL opinion tensor and the credibility tracker carry their share of the marginal accuracy gain.

4. AEGIS Architecture

4.1. End-to-end pipeline

The AEGIS pipeline ingests multi-source environmental streams, routes them through five specialist evidential

Table 1
AEGIS Specialist Agent Catalog and Modality Mappings

Agent	Input modality	Feature set	Opinion emission
Climate (C)	ERA5 total precipitation	7-day rolling precipitation sum, 30-day anomaly, soil-moisture column	Per cell-day ω_C
Satellite (S)	Sentinel-1 SAR + Sentinel-2 optical	VV, VH, VV/VH ratio, NDWI, NDVI	Per cell-day ω_S
Land-Cover (L)	ESA WorldCover + Copernicus DEM	Static class index, slope, elevation	Per pass ω_L
Air-Quality / Gauge (A)	OpenAQ PM2.5 + gauge network	PM2.5 washout signal, relative humidity, gauge precipitation	Per hour-aggregated cell-day ω_A
Document Intelligence (D)	Hydrological bulletins	SentenceTransformer embeddings of severity-tagged sentences	ω_D over bulletin-day pair

agents, fuses them through the SL coordinator, produces a calibrated prediction through a hybrid evidential head, and exposes the result through an interactive provenance dashboard. Figure 1 sketches the end-to-end architecture. Each cell-day evaluation flows from raw observation to opinion to consensus to prediction; every intermediate object is logged to the DuckDB provenance ledger for downstream auditing.

4.2. Data ingestion substrate

A canonical DuckDB 1.1 warehouse materialises nine relational tables — `climate_daily`, `satellite_sar`, `satellite_optical`, `landcover_static`, `dem_static`, `airquality_hourly`, `gauges_hourly`, `bulletins_json`, and `opinions_log` — each keyed by (`cell_id`, `timestamp`). Specialist agents read directly from this canonical store, eliminating ad-hoc file I/O and yielding a single provenance-aware substrate. Spatial cells are generated from a 200-cell regular grid over the Brisbane AOI bounding box; daily time stamps span the 24-day Brisbane 2022 flood window. The total ingested volume is 4,800 spatio-temporal instances per evaluation run, computed once per pipeline invocation and reproducible from a fixed seed.

4.3. Specialist agent catalog

Each of the five specialist agents maps a physical modality onto the shared four-state Dirichlet evidence frame. Table 1 details the specialist agent catalog.

Climate, Satellite, Land-Cover and Air-Quality agents emit a Dirichlet posterior over the four states; the Document agent emits a soft categorical verdict that is converted to a four-state opinion through the same base-rate prior for algebra compatibility. Each agent independently computes its evidence weight C and is responsible for the partial-observable projection when its modality is unavailable — a property that propagates the monotone-u invariant downstream without further coordination [Kaplan et al., 2015].

4.4. SL coordinator and routing logic

The coordinator consumes one opinion per active agent per cell-day. It first computes the maximum pairwise Jensen-Shannon divergence across active opinions using

$$\mathrm{JS}(p,q) = \frac{1}{2}D_{\mathrm{KL}}(p \,\|\, M) + \frac{1}{2}D_{\mathrm{KL}}(q \,\|\, M), \qquad M = \frac{p+q}{2} \tag{6}$$

where p and q are the projected probability vectors. The routing logic is governed by a threshold $\tau_{\mathrm{low}} = 0.1$: when the maximum pairwise JS divergence falls below τ_{low} the coordinator invokes Consensus & Compromise Fusion; otherwise it invokes Weighted Belief Fusion, weighting each active agent by its current credibility γ [van der Heijden et al., 2018, Wang and Singh, 2007]. The resulting fused opinion $\mathbf{F} = (\mathbf{b}, u, \mathbf{a})$ is the canonical object consumed by the prediction head and logged to `opinions_log`.

4.5. Hybrid evidential prediction head

The hybrid evidential head is a single LightGBM gradient-boosting classifier that consumes the 27-dimensional composite feature vector $\mathbf{F} = [\mathbf{b} \mid u \mid \mathbf{a} \mid \text{raw features}]$ constructed in Section 3.5. Two outputs are produced: a four-class flood-state prediction and a continuous flood-depth regressor sharing the same input. The composite space is essential: the 13 belief-and-uncertainty dimensions encode the consensus-level epistemic signal while the 15 raw-feature dimensions encode agent-level discriminative detail. This decomposition lets the same LightGBM fit retain a competitive accuracy surface (F1-Macro 0.7190, AUROC 0.9310) while inheriting the coordinator's calibrated uncertainty. The head is trained on all $200 \times 24 = 4,800$ cell-days under stratified cross-validation.

4.6. Provenance and explainability layer

Every agent opinion, every routing decision with the JS-divergence value and selected operator, every fused opinion, and every prediction is logged to the DuckDB provenance ledger with manifest IDs and source-level references. A FastAPI backend (mounted at port 8085) exposes the `opinions_log` table through a REST API, with security middleware enforcing a strict content-security policy. A Leaflet dark-mode spatial grid viewer interfaces with the API to render per-cell epistemic uncertainty u, per-agent credibility γ over time, and per-source contribution shares for any selected cell or day. The dashboard is the operational layer through which analysts audit which sensor or document drove a particular contribution, distinguishing stale, missing, and uncalibrated sources without post-hoc feature attribution [Hofmeister et al., 2024, Talemi et al., 2026, Lee et al., 2025, Sandeep et al., 2026, Bairam Zadeh et al., 2025].

5. Experimental Setup

5.1. Study area and event window

The evaluation is anchored on the Brisbane February–March 2022 south-east Queensland flood event, an Australian Insurance Council Class-A national disaster and a canonical benchmark for the recent agentic environmental-AI literature. The Area of Interest spans 152.5°E–153.5°E

Table 2
AEGIS Dataset Ingest Manifest and Spatial-Temporal Resolution

Dataset	Version	Spatial resolution	Temporal cadence	AOI volume
ERA5 reanalysis	ERA5-Land daily	0.1°	daily	24 daily rasters
Sentinel-1 SAR	GRD IW	10 m	6-day	50 scenes
Sentinel-2 optical	L2A	10 m	5-day	60 scenes
ESA WorldCover	v200	10 m	static	1 raster
Copernicus DEM	GLO-30	30 m	static	1 raster
OpenAQ PM2.5	real-time feed	station	hourly	8 stations × 24 d
Gauge network	BOM ALERT	station	15-min	12 stations × 24 d
Hydrological bulletins	BOM XML/JSON	AOI	daily	24 documents
EMSR + GFD v1.1 labels	Copernicus EMSR / GFD	90 m	event-day	442 polygons

and 27°S–28°S, covering the metropolitan flood-prone river corridors of the Brisbane River and its tributaries. The event window is 2022-02-20 to 2022-03-15, 24 contiguous days that include the principal precipitation exceeding the 1-in-100-year return threshold and the corresponding gradual recession.

5.2. Spatial-temporal substrate and dataset manifest

The spatial substrate is discretised into a regular grid of 200 cells × 24 daily time stamps, yielding 4,800 spatio-temporal instances per evaluation run. The dataset ingest manifest is summarised in Table 2.

5.3. Ground-truth derivation

The realised four-state label per cell-day is derived from the Copernicus Emergency Management Service Rapid Mapping and the Global Flood Database v1.1. Each cell receives a $y \in \{\text{dry, saturated, surface, inundation}\}$ class determined by an offline assignment procedure that intersects the cell with EMSR/GFD polygons and propagates class membership through the daily inset map. Cells outside any observed inundation polygon for a given day are labelled dry. Bordering cells close to observed inundation but absent from active polygons are conservatively labelled saturated. The 442 EMSR/GFD polygons cleanly identify the realised inundation timeline [Hofmeister et al., 2023, 2024].

5.4. Baselines

Four methods are evaluated on the same 200×24 cell-day fold. AEGIS-SL is the full proposed framework. BL1 is an ERA5-only LightGBM filter that excludes Sentinel, Land-Cover, Air-Quality, and Document inputs and uses only reanalysis climate features. BL2 is a monolithic late-fusion LightGBM that concatenates the 15 raw-feature channels from all five agent sources and trains on a 27-dimensional raw-feature vector excluding the 12 opinion dimensions and the u mass. BL3 is an LLM-arbitrated system that prompts a Qwen2.5-3B-Instruct model (running via llama.cpp 4-bit local) with each agent's verdict summary plus the raw modal features and asks the model to emit a four-state class decision. The LLM baseline serves as the contemporary agentic state of the art and is the primary foil against which AEGIS-SL is benchmarked.

5.5. Metrics

The four reported metrics are: Macro-F1, Weighted-F1, AUROC (one-vs-rest macro), and Expected Calibration Error computed with 10 equal-width probability bins. Reported ECE values are for the calibrated probabilistic outputs only; baselines that emit non-probabilistic decisions report a minus sentinel for ECE. Two operational metrics are also logged for completeness: wall-clock latency per pipeline invocation, and RAM peak from Python tracemalloc. The Brier-score reputation updates per agent are logged per cell-day for the credibility ablation analysis in Section 6.6.

5.6. Implementation and reproducibility

The implementation is in Python 3.11 with DuckDB 1.1 as the canonical warehouse, NumPy 1.26 and PyTorch 2.3, LightGBM 4.3, and llama.cpp for the local Qwen2.5-3B-Instruct 4-bit model. All experiments execute on a single 12-core CPU node. SL operator algebra is implemented from scratch using NumPy, independently cross-validated against an external Subjective Logic reference implementation for operator correctness on 200 randomly seeded Dirichlet samples (200/200 pass) [van der Heijden et al., 2018]. The full test suite comprises 34 unit tests (34/34 pass) covering operator correctness, projection monotonicity, fusion associativity, and dashboard serializer output. The complete implementation, datasets, evaluation harness, and dashboard are open-source and publicly accessible[2].

6. Results

6.1. Hypothesis summary

Four pre-registered hypotheses were evaluated on the same 200-cell × 24-day substrate (4,800 cell-day instances) on a single 12-core CPU node. H1 asks whether the AEGIS-SL full pipeline matches the monolithic raw-feature late-fusion baseline under equal data and compute. H2 asks whether LLM-arbitrated arbitration, the contemporary state

[2] https://github.com/ali-Hamza817/AEGIS

Table 3
Empirical Benchmark Performance Comparison Across Methods

Method	F1-Macro	F1-Weighted	AUROC (ovr)	ECE (10-bin)	Latency (s)	RAM peak (MB)
AEGIS-SL (full)	**0.7190**	**0.7240**	**0.9310**	**0.043**	14.7	612
BL2 — monolithic raw-feature late fusion	0.7106	0.7150	0.9189	0.087	11.2	588
BL1 — ERA5-only LightGBM	0.5880	0.6210	0.7820	0.214	2.1	248
BL3 — LLM-arbitrated (Qwen2.5-3B-Instruct 4-bit)	0.6238	0.6891	0.8410	—	86.4	1,820

Table 4
BL3 Minority-Class Collapse Diagnostic

Class	BL3 precision	BL3 recall	BL2 recall	AEGIS-SL recall
dry	0.92	0.94	0.96	0.95
saturated	0.84	0.21	0.58	0.61
surface	0.71	0.09	0.49	0.52
inundation	0.89	0.18	0.71	0.73

of the art, is structurally vulnerable to minority-class collapse. H3 asks whether epistemic uncertainty u rises monotonically as modalities drop. H4 asks whether the provenance dashboard supports source-level auditable contributions. Section 6.6 reports a mechanistic 2×2 ablation isolating the marginal contribution of the SL opinion tensor and the credibility tracker.

6.2. H1 — Benchmark comparison

Table 3 presents the master benchmark comparison across all four methods evaluated on the same fold.

AEGIS-SL attains the highest F1-Macro, highest AUROC, and lowest ECE on the same evaluation fold. The margin over BL2 — the strongest alternative — is statistically modest in F1 (+0.0084) but substantial in calibration (ECE 0.043 vs 0.087, a 50.6% improvement). BL1 trails because reanalysis precipitation alone cannot resolve fine-scale surface inundation patterns; BL3 trails because LLM-arbitration incurs a $\sim 6\times$ wall-clock cost relative to AEGIS-SL while producing worse majority-class precision, and the ECE column is reported as a null since BL3 emits non-probabilistic decisions. These results support H1.

6.3. H2 — BL3 minority-class collapse diagnostic

Table 4 details the class-wise precision and recall breakdown across BL3, BL2, and AEGIS-SL.

The LLM-arbitrated baseline exhibits the precise failure mode predicted by the structural analysis in Section 1 — moderate-to-strong ranking quality on the majority class but severe collapse on the minority classes that operationally matter most: saturated (0.21 recall), surface (0.09), and inundation (0.18). The error is not random; it is systematic suppression. Both BL2 and AEGIS-SL preserve substantially higher minority-class recall, with AEGIS-SL retaining a small but consistent edge over BL2 across all three minority classes. These results support H2 and confirm that prompt-chained arbitration is structurally vulnerable to confident-but-wrong majority-class skew for the classes that drive evacuation decisions.

Table 5
Monotonic Epistemic Uncertainty u Under Modality Dropout

Modalities dropped	Mean u	Median u	Min u	Max u	Std u
0 (full)	0.12	0.11	0.03	0.41	0.06
1	0.34	0.33	0.18	0.58	0.09
2	0.61	0.62	0.41	0.84	0.10
3	0.83	0.85	0.66	1.00	0.09

Table 6
24-Day Mean Per-Agent Credibility Trajectory and Reputation

Agent	Final γ (24-day mean)	Initial γ_0	Brier score mean
Satellite (S)	0.91	0.50	0.14
Climate (C)	0.86	0.50	0.17
Document (D)	0.81	0.50	0.20
Land-Cover (L)	0.78	0.50	0.22
Air-Quality / Gauge (A)	0.74	0.50	0.24

6.4. H3 — Monotonic-u progression under modality dropout

Table 5 reports the distribution of epistemic uncertainty u as modalities are systematically removed.

Mean u rises monotonically from 0.12 to 0.83 across the progression $0 \to 1 \to 2 \to 3$ dropped modalities; the median tracks the mean within 0.02 across all four levels; the minimum-u floor also rises ($0.03 \to 0.66$) confirming that every cell becomes more uncertain under dropout, never fewer. No dropouts in the evaluation fold produced a non-monotone u value, satisfying the partial-observable projection invariant from Section 3.2 [Kaplan et al., 2015]. These results support H3 and demonstrate that missing-modality events can be detected through a monotonic u increase logged to the provenance ledger — a property that distinguishes AEGIS from the surveyed agentic pipelines which silently recover or hallucinate under dropout.

6.5. H4 — Provenance, credibility, and per-source contribution

Per-cell provenance queries issued through the dashboard return the full manifest: agent identity, evidence weight C, base-rate $\mathbf{a}$, SL coordinator's measured Jensen–Shannon divergence value, the operator selected, the resulting fused $\mathbf{b}/u/\mathbf{a}$ tuple, and the LightGBM head's 27-dimensional feature snapshot at evaluation time. The 24-day mean per-agent credibility trajectory converges within the first seven days to stable final reputation values, as reported in Table 6.

Per-source contribution to the fused opinion, summarised as mean share of evidence weight across the full dataset,

Table 7
Mechanistic 2×2 Ablation of AEGIS-SL Components

Configuration	SL opinion tensor	Credibility γ	F1-Macro	AUROC	ECE
Full AEGIS-SL	ON	ON	**0.7190**	**0.9310**	**0.043**
SL only	ON	OFF	0.7051	0.9210	0.058
Credibility only	OFF	ON	0.6988	0.9142	0.082
Neither (BL2-like)	OFF	OFF	0.6678	0.8980	0.094

places satellite imagery (41.2%) and climate forecasts (28.4%) as the dominant contributors, with land-cover (11.5%), air-quality (9.7%), and document intelligence (9.2%) acting as listening-secondarily corroborators. Importantly, when an agent drops a modality in the H3 progression, contribution does not silently redistribute: the partial projection u expansion is recorded and the unreliable agent's reputation decays through γ tracking rather than being silently zeroed out [Wang and Singh, 2007]. These results support H4 and demonstrate that an analyst can audit which sensor or document drove each contribution and detect which source was stale, missing, or uncalibrated after the fact.

6.6. Mechanistic 2×2 ablation

To isolate where the marginal accuracy comes from, we run a 2×2 ablation on the AEGIS-SL framework by toggling whether the SL opinion tensor $(\mathbf{b}, u, \mathbf{a})$ is fed into the 27-dimensional composite feature space and whether the credibility tracker γ is active in WBF weighting. Table 7 summarizes the configuration results.

The SL opinion tensor carries the larger marginal contribution: enabling SL alone yields +0.0373 F1-Macro and +0.0230 AUROC over the BL2-like baseline, while enabling credibility alone yields +0.0310 F1-Macro and +0.0162 AUROC. The two components are synergistic — their joint activation produces the highest accuracy and lowest calibration error, exceeding either component in isolation. Removing both components collapses the framework to monolithic late-fusion performance (F1-Macro 0.6678, AUROC 0.8980), providing internal evidence that the methodological contribution of the AEGIS framework is structurally attributed to the evidential machinery and not to an incidental architectural detail [van der Heijden et al., 2018, Wang and Singh, 2007, Hofmeister et al., 2024].

7. Conclusion

7.1. Summary

AEGIS demonstrates that the four structural gaps blocking operational deployment of agentic environmental AI — calibrated minority-class handling, monotonic uncertainty under missing modalities, propagating per-agent credibility, and source-level provenance — can be addressed in a single, formally grounded framework. The pipeline fuses five specialist agents over a shared four-state Dirichlet frame through formal Subjective Logic operators, propagates per-agent reputation through a Brier-score-credited trust model, and surfaces a per-cell provenance manifest through a DuckDB-backed interactive dashboard. On the Brisbane February–March 2022 case study, AEGIS-SL attains the highest F1-Macro, AUROC, and lowest calibration error across all four evaluated baselines while uniquely providing the four properties absent from the surveyed agentic pipelines.

7.2. Limitations

The framework is currently single-event: the evaluation fold is anchored on Brisbane 2022, and per-event knowledge-graph context is held out for future work. The LLM arbitrator baseline was restricted to a 3-billion-parameter 4-bit local model for reproducibility, leaving open whether a stronger proprietary LLM could close the minority-class gap through pure scale. Fusion operators remain deterministic and do not yet expose a learning-based meta-policy over the WBF / CCF routing threshold τ. Lastly, the dashboard is rich for a single study area but has not been stress-tested across multiple concurrent AOIs.

7.3. Broader applicability

The AEGIS architecture is event-agnostic by design. The five specialist agents — climate, satellite, land-cover, air-quality, and document intelligence — and the four-state Dirichlet frame are domain-portable: wildfires, landslides, heat domes, and air-quality crises can each be reformulated as a four- or five-state progression through the same specialist-agent catalog and operator algebra. The provenance ledger is also agent- and modality-agnostic, which makes the same dashboard operational layer a candidate for shared deployment across a federation of environmental risk domains. Per-event knowledge graphs from the Hofmeister et al. [2024] semantic-agent line remain the natural extensibility surface, suggesting that the AEGIS contribution is best understood as a formal evidential extension of an existing agentic paradigm rather than a competing architecture [Talemi et al., 2026, van der Heijden et al., 2018].

7.4. Future work

Three concrete extensions are natural next steps. First, multi-event generalisation requires re-running H1–H4 on a Christchurch 2011 or Germany 2021 AHR flood fold to test whether the credibility tracker converges within seven days across heterogeneous events. Second, a learning-based routing policy could replace the fixed $\tau_{\text{low}} = 0.1$ cutoff with a small neural router trained on operational analyst preferences. Third, foundation-model adapters for the Document Intelligence agent would let bulletin embeddings absorb unsupervised linguistic drift across agencies. We make the full implementation publicly available to support independent replication and extension across all three directions[3].

[3] https://github.com/ali-Hamza817/AEGIS

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